

System Design: Ticket Booking Platform

This document outlines the architecture, concurrency control, and logic flows powering the high-throughput seat booking and waitlist mechanisms. The platform is designed around PostgreSQL (Supabase) to strictly enforce transactional integrity and prevent race conditions.

1. Concurrency Prevention & Isolation

High-demand events are notorious for concurrency issues multiple users attempting to book the exact same seat simultaneously.

To solve this, the platform utilizes strict database-level locking and status isolation instead of relying on frontend validation.

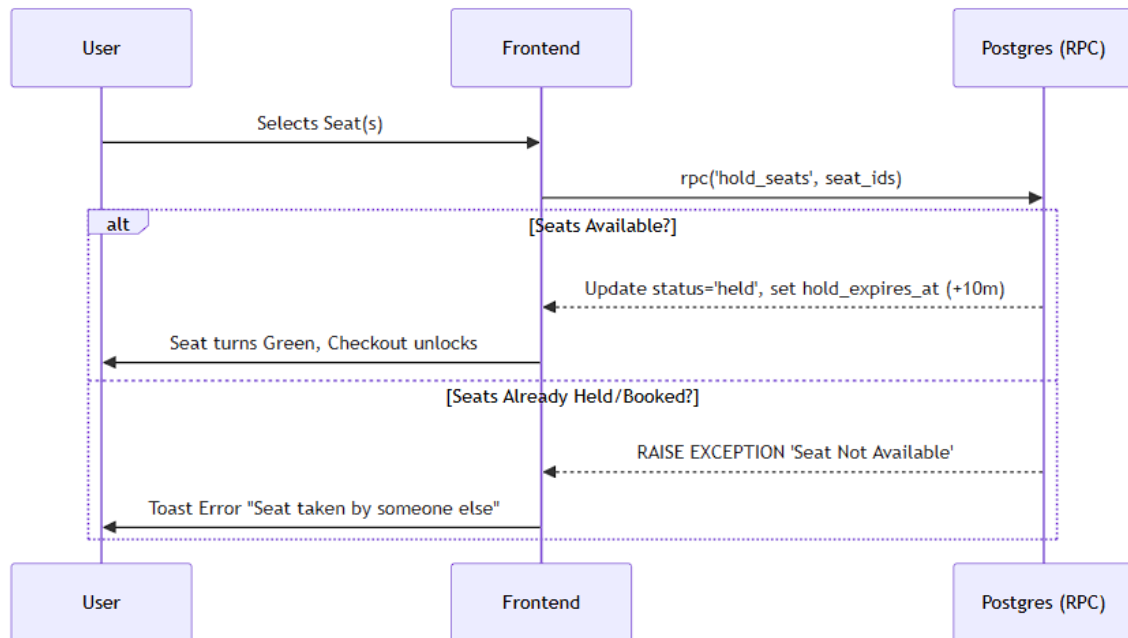
- All critical mutations happen via Postgres Functions (RPCs) which run inside atomic transactions.
- We rely heavily on the FOR UPDATE and FOR UPDATE SKIP LOCKED clauses.
- SKIP LOCKED: Essential for the waitlist assignment flow. When looping over cancelled seats and querying for the next waiting customer, SKIP LOCKED guarantees that if multiple concurrent transactions are processing waitlists, they will seamlessly skip over rows that are already being processed by another transaction, entirely avoiding deadlocks and double-offers.

2. Seat Hold and TTL Mechanism

Instead of requiring upfront payment to reserve a seat, we use a Time-To-Live (TTL) soft-hold mechanism. This dramatically improves user experience by securing their selection while they navigate the checkout flow.

Architecture

- The `show_seats` table acts as the source of truth for seat states.
- It uses three primary statuses: available, held, and booked.
- The `hold_seats(seat_ids)` RPC updates selected seats to held, sets `held_by = auth.uid()`, and injects a TTL via `hold_expires_at = now() + 10 minutes`.

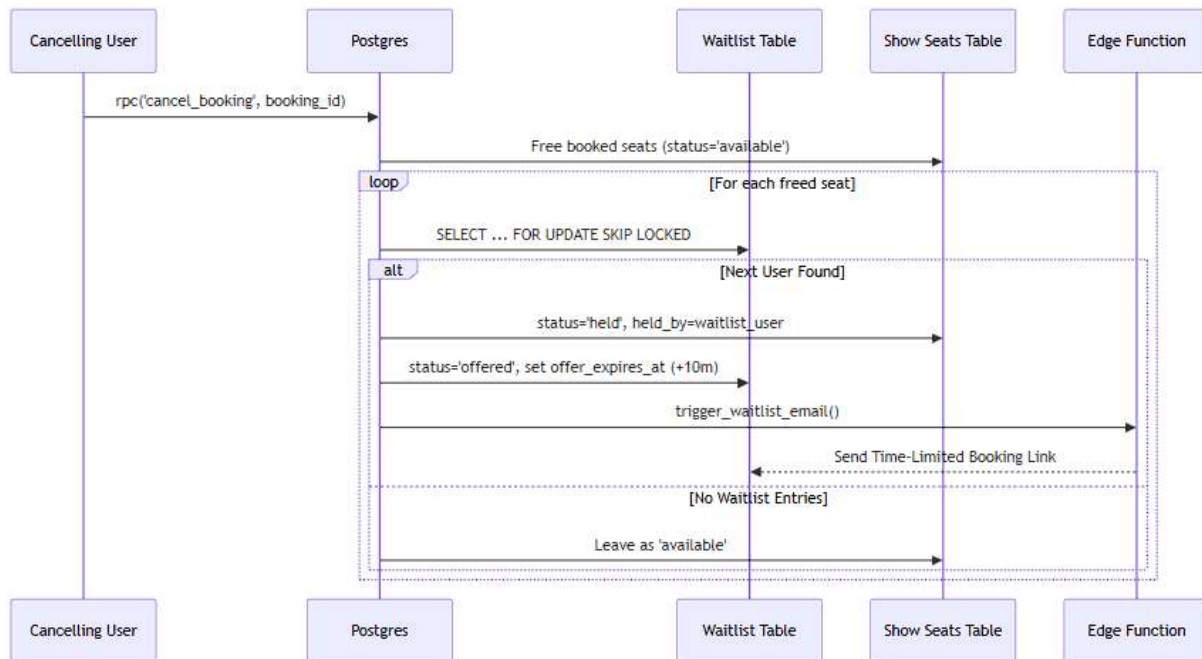
Hold Flow Diagram**3. Waitlist Queue & Auto-Assignment Flow**

When a seating category hits capacity (available = 0), the UI switches to “Sold Out” and unlocks the Waitlist capability.

Ticket Quantity Support

Users can join the waitlist and request multiple tickets. Rather than grouping these into a single complex database row, the `join_waitlist` RPC inserts one queue row per requested ticket. This guarantees granular, one-to-one mapping between freed seats and waitlist progression, preventing situations where a user who wants 4 tickets permanently blocks the line because only 2 seats keep cancelling.

Waitlist Flow Diagram



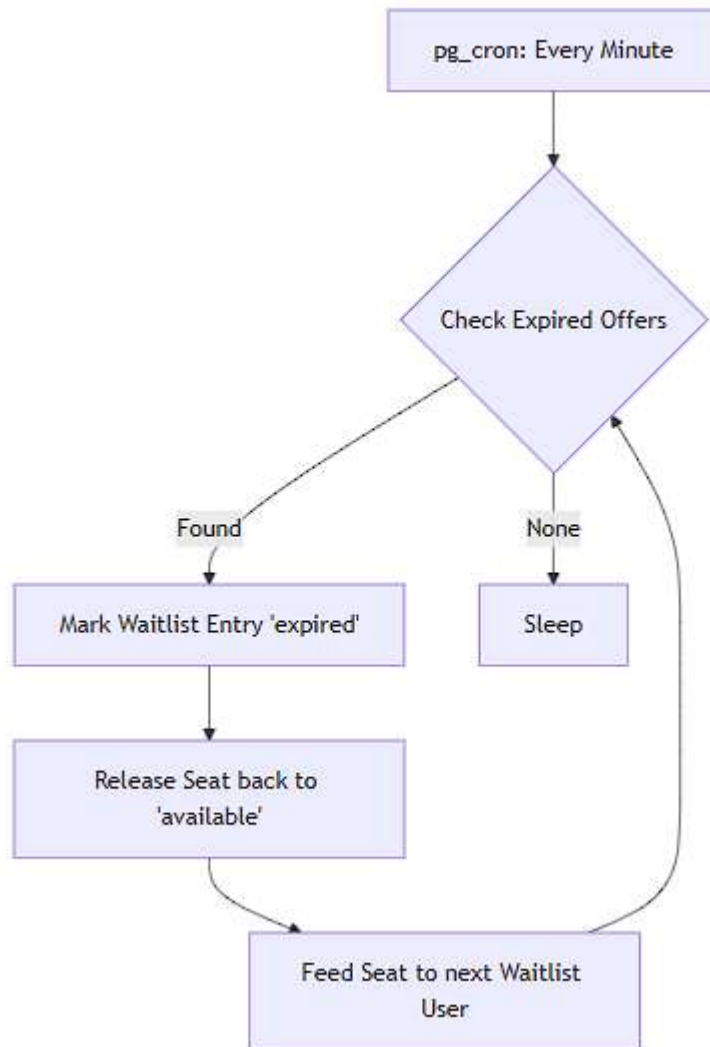
4. Time-Limited Offer Handling

Waitlist offers must be strictly time-boxed to ensure that if a user is asleep or ignores the email, the seat doesn't remain deadlocked forever.

The Lifecycle of an Offer

1. Offer Sent: The seat is hard-reserved (held) specifically for the waitlisted user's uid. The waitlist row is updated to status = 'offered' with a 10-minute expiration.
2. Acceptance: If the user clicks the email link, they call `accept_waitlist_offer()`. This transitions the waitlist entry to fulfilled and resets the seat's `hold_expires_at` for another 10 minutes, giving them time to process payment.
3. Expiration: We use `pg_cron` running every 60 seconds (`expire-waitlist-offers-cron`). It scans for `offer_expires_at <= now()`. It transitions expired offers to expired status and instantly loops the abandoned seat back into `process_waitlist_for_seat()`, bumping it to the *next* person in line automatically without manual admin intervention.

Expiration Loop Diagram



Note on Edge Case Handling: When a waitlist offer expires, it is critical that the seat is transitioned strictly back to available *before* invoking `process_waitlist_for_seat()`. If the seat remains held by the expired user, the assignment logic will fail to pick it up, resulting in a deadlock where the seat sits empty on the floor but the next user in line is stranded on the waitlist. Our SQL functions strictly enforce this sequence

5. QR Code Generation and Delivery

Upon final booking confirmation, the system must securely issue tickets to the user.

Delivery Architecture

1. Trigger: The frontend finalizes the payment/booking via the `book\_held\_seats` RPC. Upon receiving a success response, it triggers the `/api/send-ticket-email` Next.js API route.
2. Payload Generation: The API route queries the database for the user's new booking record. It uses the `qrcode` library to generate a base64 Data URI representing the ticket's booking reference and user ID.
3. Dispatch: We use Resend (via `@react-email`) to inject the generated QR code payload into a styled HTML React component. The email is asynchronously dispatched to the user's registered Supabase Auth email address, delivering a high-quality, scannable digital ticket directly to their inbox.